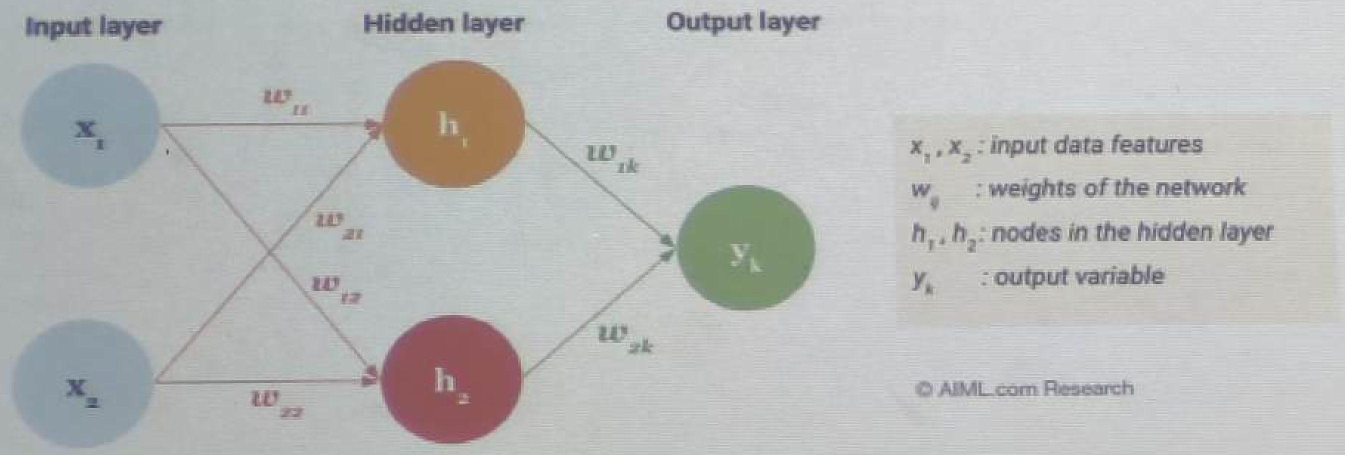
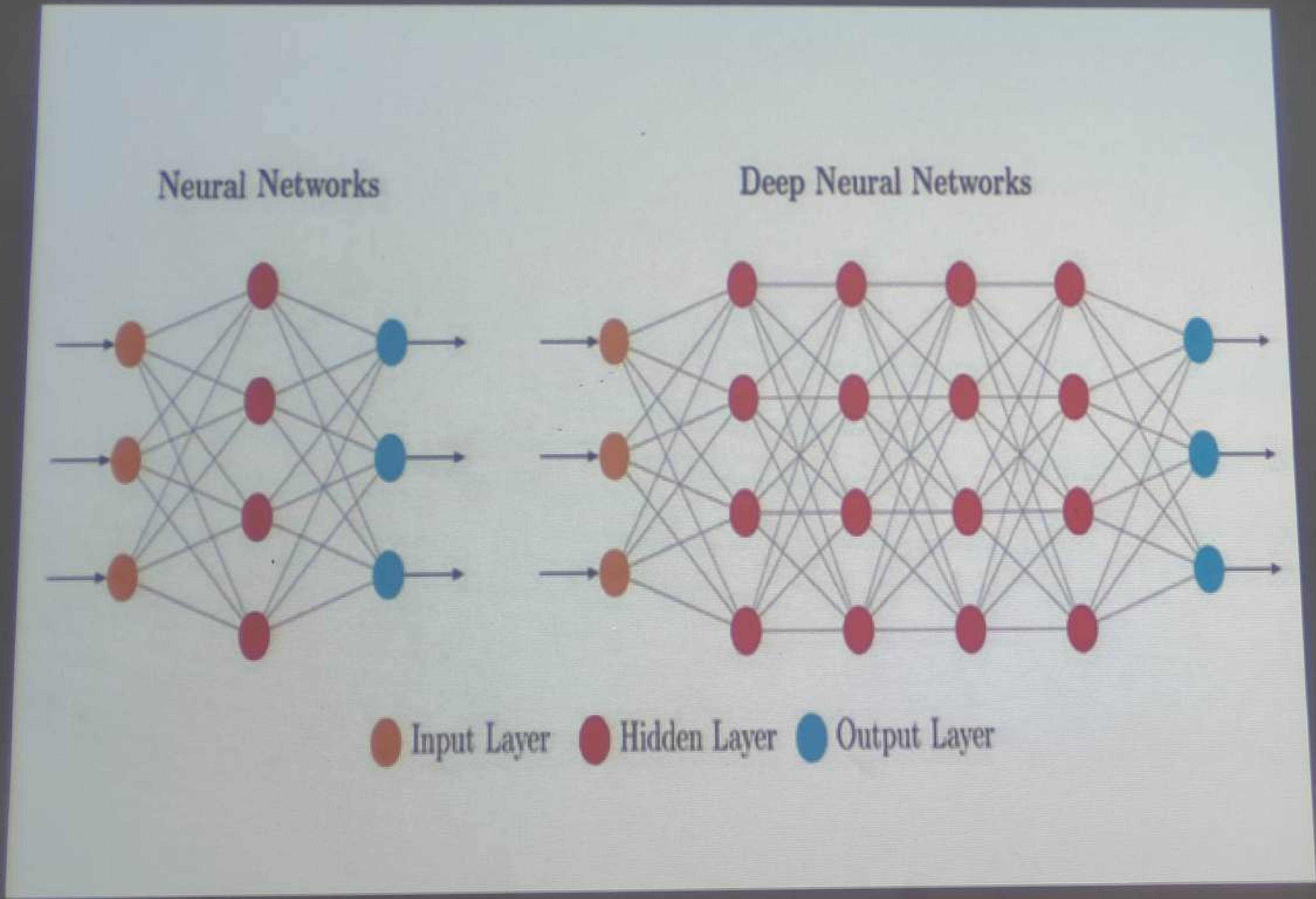
Artificial Neural Network

- Artificial Neural Network is a machine learning model consists of network of interconnected neurons inspired by biological neurons.
- Neurons are the fundamental processing units. Each neuron receives inputs, performs a calculation (typically a weighted sum and activation function), and produces an output.



This example consists of an input layer with two input nodes, one hidden layer with two neurons and an output layer with one neuron


Neural Networks
Deep Neural Networks
Input Layer
Hidden Layer
Output Layer

Attention Mechanisms

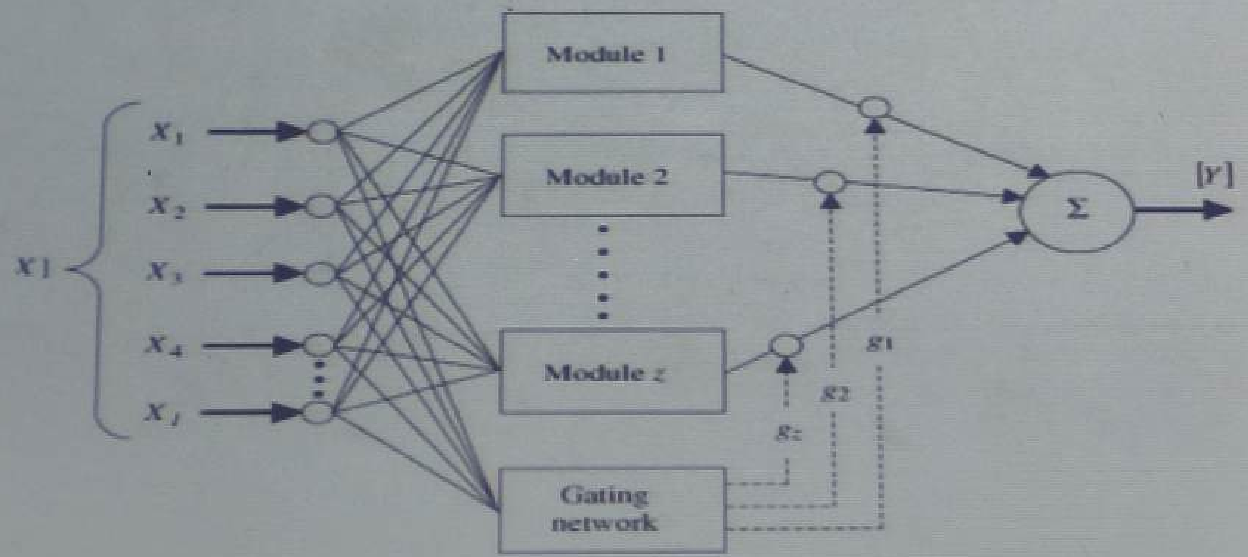
- Attention mechanisms is a technique in neural network/model which allows it to focus on the most important parts of the input while making predictions.
- It behaves like "Instead of looking at everything equally, focus more on what's relevant.“
- This is achieved by assigning weights to different parts of the input, with the most important parts receiving higher weights.
- This mechanism helps improve the model's performance by allowing it to prioritize relevant information and disregard irrelevant noise.

Attention Mechanisms

- Ex. Imagine you're reading a paragraph to answer a question:
  "What is the capital of France?" You don't read every word equally — your eyes search for "capital" and "France", then focus on "Paris". This is exactly what an attention mechanism helps a model do.
- Why Attention is Useful in Explainability? Because it naturally gives us insight into: 1. Which parts of the input the model considered important. 2. How much "attention" was given to each input token/image patch which helps explain why the model made a decision.

Modular Networks

- A Modular Neural Network (MNN) is a type of AI architecture where the network is divided into smaller independent "modules" — each responsible for solving a specific part of the overall task.
- A Modular Network breaks down a complex problem into smaller, specialized sub-networks (modules), which makes the entire system easier to understand, manage, and explain.



Modular Networks

Ex. Self-Driving Car

Module	Task
Vision Module	Detects lanes, traffic signs
Navigation Module	Plans route
Control Module	Steers and accelerates car
Safety Module	Takes emergency actions

Each module works independently, but all contribute to the final task

Modular Networks

Benefit	Explanation
Interpretability	Each module does a well-defined sub-task → easier to understand
Debugging	Easier to identify which part failed or behaved wrongly
Efficiency & Scalability	Different modules can be reused or trained in parallel
Flexibility	You can swap modules without redesigning the whole system
Task Specialization	Each module becomes an expert in its own subdomain

Feature Identification

- Feature identification refers to the process of determining which input features (variables) are most important or influential in a machine learning model's prediction.
- Feature identification makes black-box models more explainable by helping us answer: Why did the model make this prediction? What were the top contributing factors?
- Integrated Gradients, Saliency Maps, Grad-CAM are the tools for fearture identification for deep learning.
- Ex. If the model rejects a loan, feature identification may show: Credit Score = 0.45 importance

  Income = 0.30

  Loan Amount = 0.20

  Now we know: Credit Score was the most critical factor.

Feature Identification

Benefit	Explanation
Explainability	Know why the model made a decision
Debugging	Identify if the model is relying on wrong features
Fairness	Detect if decisions are biased against certain inputs
Model Simplification	Can remove less important features

Feature Visualization

- Feature Visualization refers to techniques used to understand what a neural network has learned by visualizing the internal features (patterns, shapes, concepts) that activate certain neurons.
- It answers, "What kind of input activates this neuron or layer the most?"
- Ex. Imagine training a dog to recognize cats and dogs. If you could peek inside its brain, you might see: Floppy ears → dog and Whiskers & triangle ears → cat. For a CNN trained to detect animals: Early layers → detect edges, colors, textures, Middle layers → detect shapes, fur patterns and Deeper layers → detect whole faces or paws.
- It helps identify which patterns or textures are being picked up and also verifies if model is learning relevant concepts (like "fur" or "eyes")

Grad-CAM

- Grad-CAM stands for Gradient-weighted Class Activation Mapping.
- It is a technique used in deep learning, particularly with Convolutional Neural Networks (CNNs), to visualize which parts of an image are most important for a model's prediction.
- It helps explain the "why" behind a CNN's decision by generating a heatmap highlighting important regions.
- Essentially, it identifies the areas of the input that significantly influence the final classification score.

Grad-CAM

Original Image | Grad-CAM 'Cat' | Grad-CAM 'Dog'

Grad-CAM

How Grad-CAM Works?

1. Input an image into a CNN (e.g., ResNet)
2. The network makes a prediction (e.g., "cat")
3. Grad-CAM computes the gradient of the predicted class with respect to the final convolutional layer
4. It weights the feature maps(Activation Maps) by the gradient values
5. Combines the feature maps into a heatmap (class activation map)
6. Overlay the heatmap on the original image → You get where the model looked

Sensitivity Analysis

- Sensitivity analysis in neural networks is a technique to understand how changes in input features or model parameters affect the network's output.
- In simple terms: "If I slightly change this input feature, how much does the output change?"
- Ex. Let's say you have a model that predicts loan approval based on: Income, Credit Score, Age
- Now you slightly increase income from ₹50,000 to ₹51,000. If prediction changes drastically-> Model is highly sensitive to income, If prediction barely changes-> Model is robust to income changes
- Sensitivity Analysis is a simple yet powerful tool to probe the behavior of machine learning models. It helps determine which inputs matter most, and whether the model's behavior is stable and trustworthy.

Sensitivity Analysis

There are two types:

1. Local Sensitivity Analysis
   - Change one feature at a time
   - Observe the change in prediction
   - Focuses on a specific input instance
2. Global Sensitivity Analysis
   - Vary all input features over their entire range
   - Measure impact on output across many examples

Ante-hoc Explainability (AHE) Models

Examples of Ante-hoc Models:

Model Type	Why It's Explainable
Decision Trees	Easy to trace path from root to leaf
Rule-based Models	If-then logic is transparent
Linear/Logistic Regression	Coefficients show the weight of each feature
Naive Bayes	Probabilistic reasoning with visible contributions

Ante-hoc Explainability (AHE) Models

- Ante-hoc explainability refers to models that are inherently interpretable by design — meaning their structure, logic, and predictions are explainable from the beginning (before deployment).
- "Ante-hoc" = Before the decision is made and these models don't need extra explanation tools — they explain themselves naturally.
- Ante-hoc explainable models are transparent by design and help users, regulators, and developers trust and interpret decisions without needing extra tools.

Ante-hoc Explainability (AHE) Models

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Post-hoc Explainability (PHE) Models

- Post-hoc explainability refers to methods that are applied after a machine learning model is trained to explain how or why it made a certain decision.
- These techniques are especially useful when working with black-box models like neural networks, ensemble models, etc.
- Post-hoc explainability models allow us to extract insightful, visual, or numeric explanations from already trained black-box models, balancing accuracy with interpretability.

Post-hoc Explainability (PHE) Models

Examples of Post-hoc Explainability Techniques:

Method/Tool	What It Does
LIME	Builds simple surrogate models locally to explain output
SHAP	Assigns contribution values to each feature
Grad-CAM	Highlights image regions that influenced prediction
Saliency Maps	Shows which input pixels were most influential
Feature Importance	Ranks input features by their global influence

Comparison

Aspect	Ante-hoc	Post-hoc
Time of Explanation	Before prediction (built-in)	After prediction (external tools)
Model Transparency	High	Low (black-box)
Flexibility	Less (simple models)	More (any complex model)
Accuracy	Often lower	High-performing models supported
Examples	Logistic Regression, Decision Trees	SHAP, LIME, Grad-CAM

Interactive Machine Learning (IML)

- Interactive Machine Learning (IML) is a human-in-the-loop approach where users interact with the ML model during training or prediction to provide feedback, make corrections, or guide the learning process.
- IML = AI + Human Collaboration
- Ex. Think of Google Photos suggesting tags like "dog" or "wedding."If you correct a wrong tag — the system learns from that feedback.
- Interactive Machine Learning (IML) bridges the gap between human intuition and machine intelligence. It enhances explainability by letting users guide, adjust, and understand model behavior.

Interactive Machine Learning (IML)

IML Steps:

1. Model makes prediction
2. User evaluates/explains/corrects it
3. Model updates or retrains based on feedback
4. Cycle repeats

Benefits:

Better accuracy

More trust

Higher interpretability

BETA (Black-box Explanation Through Transparent Approximation)

- BETA (Black-box Explanation through Transparent Approximation) is an approach where a complex, opaque (black-box) machine learning model is approximated by a simpler, interpretable model — so that humans can understand and trust its behavior.
- Many modern ML models (like deep neural networks, ensemble methods, etc.) offer high accuracy but low interpretability. BETA helps bridge this gap by building an interpretable approximation (like decision trees, rules, or linear models) that behaves similarly to the black-box model, at least locally or on average.
- Ex. Suppose a black-box model predicts credit approval.

  Decision Rule (from surrogate):

  If credit score > 700 and income > ₹50,000 → Approve

  Else if credit score < 650 → Reject

  Else → Further review needed
- Even if the original model is a neural network, you now have transparent logic!

Hybrid Models

- Hybrid Models in XAI are systems that aim to combine the strengths of: Highly accurate but black-box models (like deep neural networks, ensemble methods) AND Transparent, interpretable models (like decision trees, rule-based systems)
- The goal of Hybrid Models is to achieve both high accuracy and good interpretability by designing systems that use both types of models intelligently.
- Hybrid models try to offer a middle ground — accuracy with insight.
- Types:
  - Rule + Deep Learning: Train neural networks but use rules to guide or filter predictions
  - White-box Inside Black-box: Use interpretable models as subcomponents inside a deep model (e.g., attention modules)
  - Black-box + Surrogate: Wrap a black-box with a local surrogate model that explains it (e.g., SHAP + DNN)
  - Human-in-the-loop Hybrids: Combine interactive feedback (IML) + traditional models